

File: cm5_socket.kicad_sch

File: system_mcu.kicad_sch

File: ac_dc_power.kicad_sch

File: digital_io_fan_encoder.kicad_sch

File: usb3_hub.kicad_sch

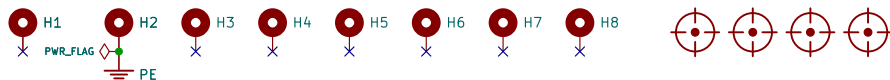
File: usb_conn.kicad_sch

File: video_outputs.kicad_sch

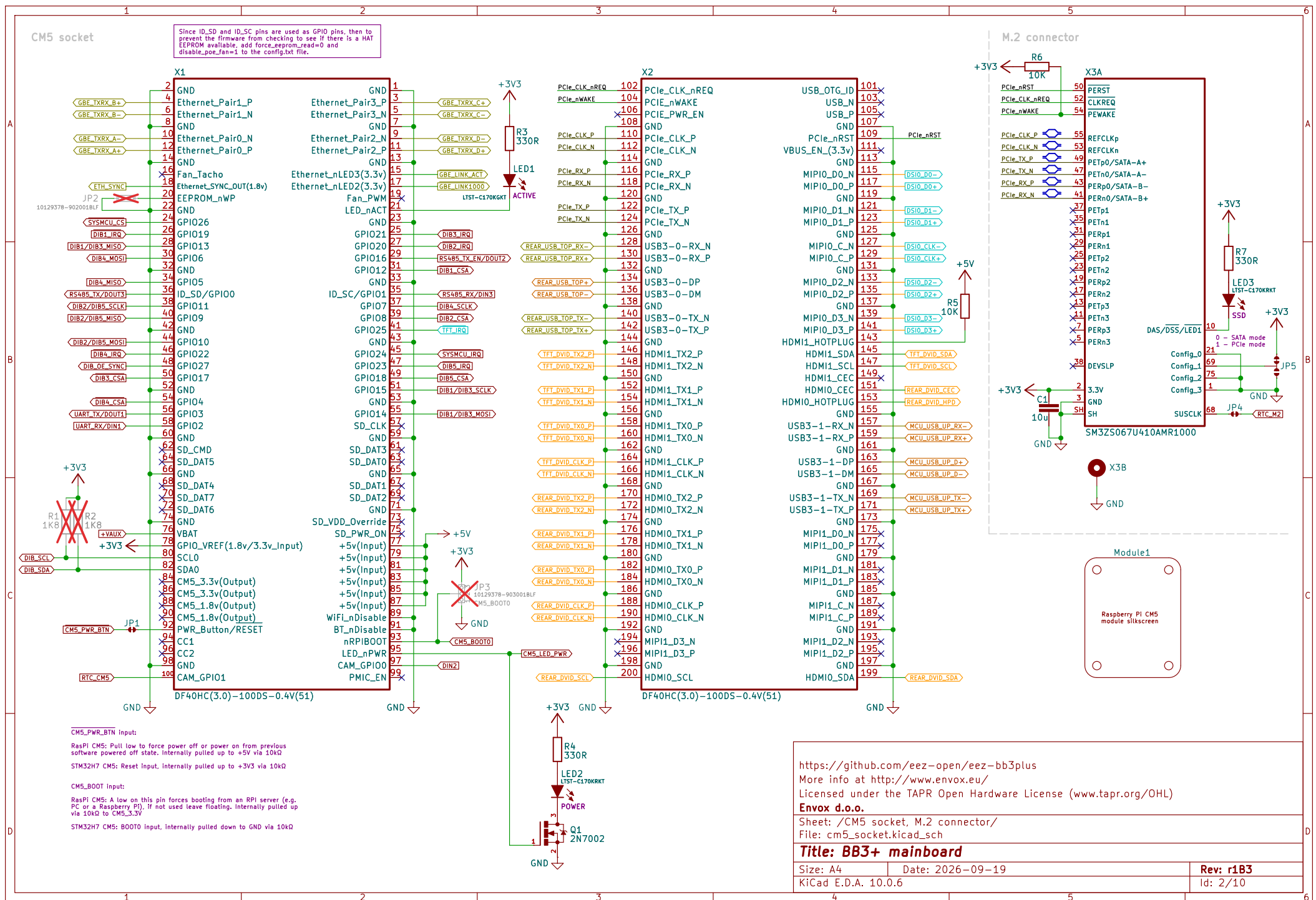
File: dib.kicad_sch

File: power_coupling.kicad_sch

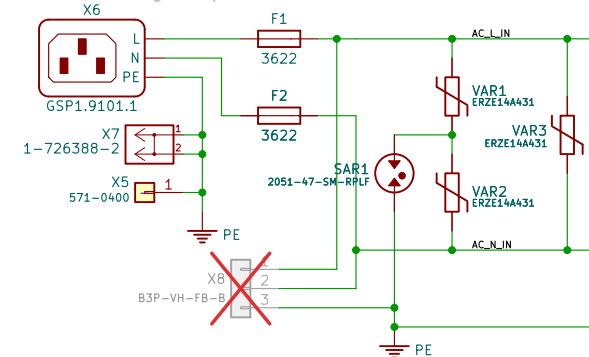
I2C address	Device
0x51	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x7A	PTN5150A USB CC controller



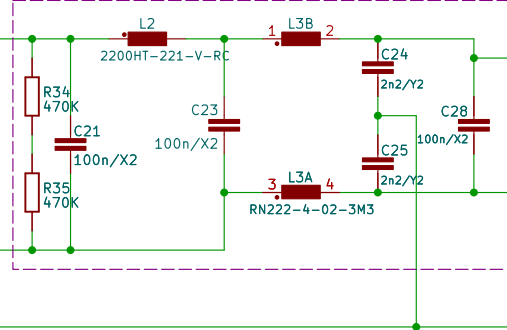
Id: 1/10



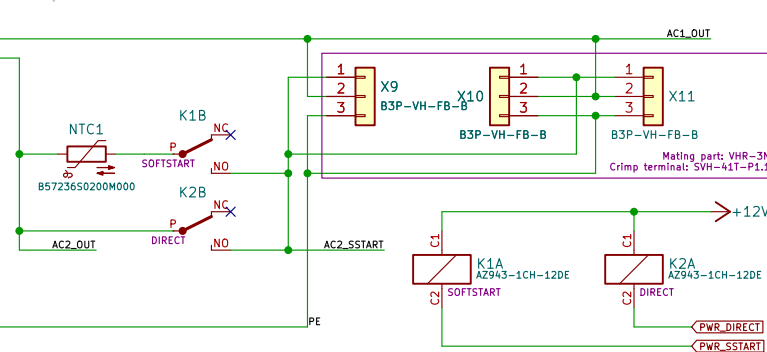
AC inlet, filtering and protection



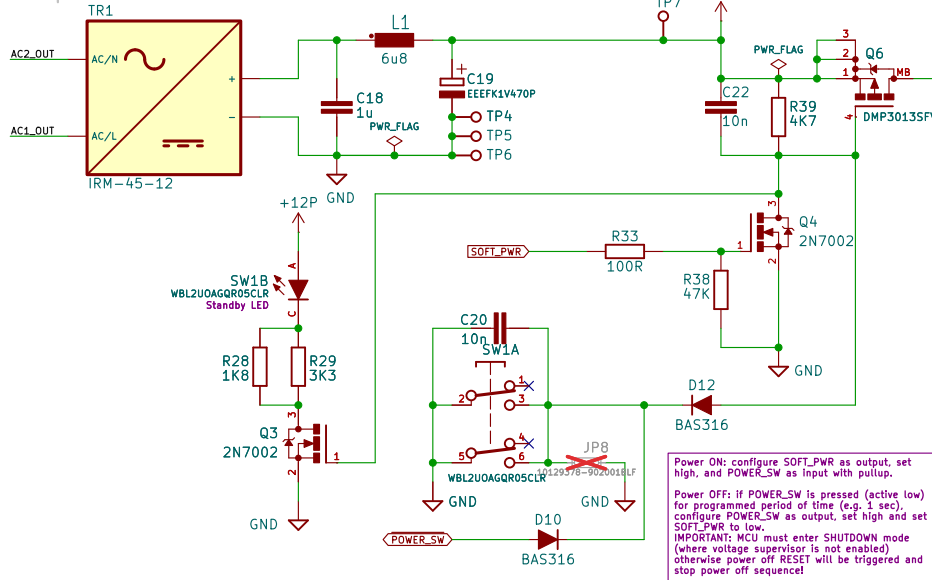
EMI compliance filter



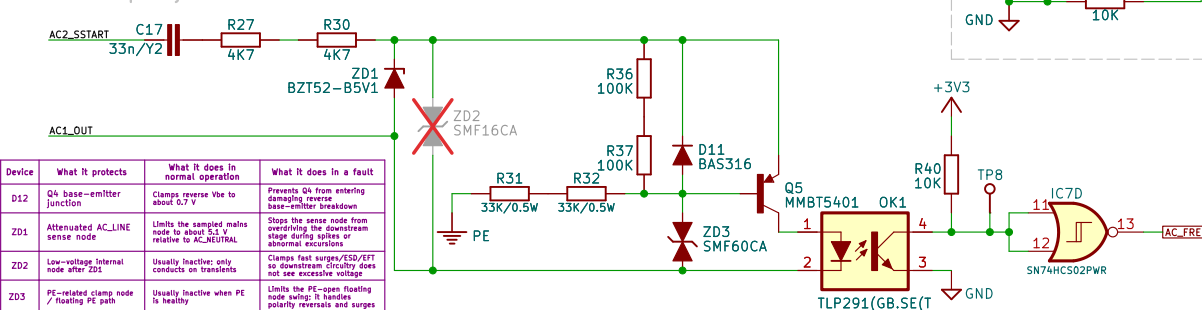
AC power control, Inrush current limiter



Soft power switch

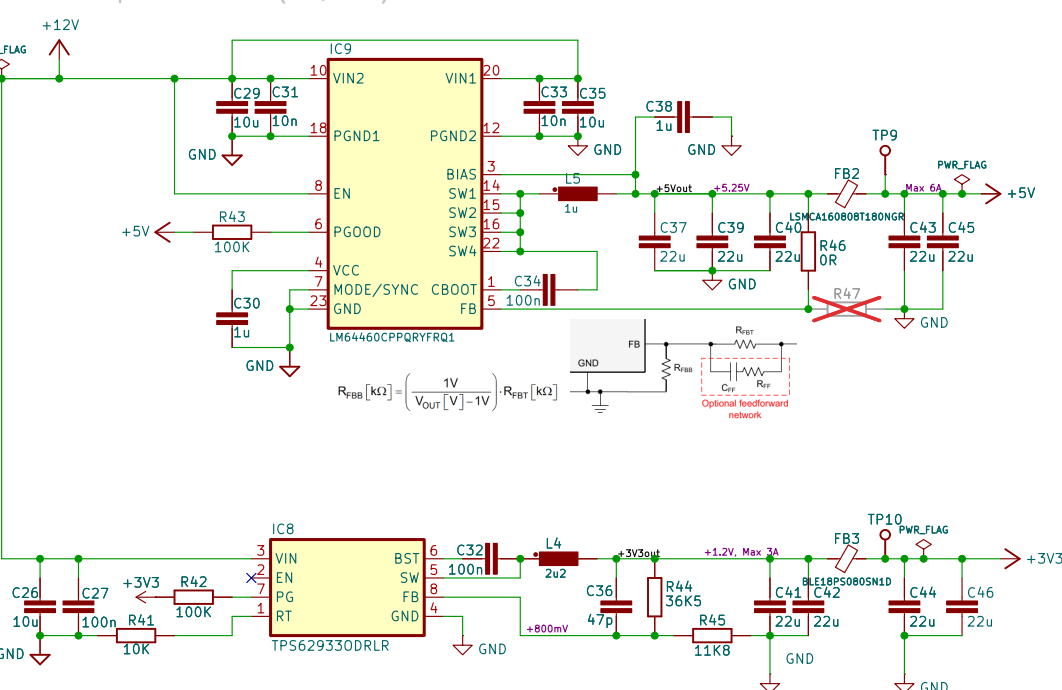


AC Mains frequency and PE fault detection



Device	What it protects	What it does in normal operation	What it does in a fault
D12	Q4 base-emitter junction	Clamps reverse Vbe to about 0.7 V	Prevents Q4 from entering damaging reverse base-emitter breakdown
ZD1	Attenuated AC_LINE sense node	Limits the sampled mains node to about 5.1 V relative to AC_NUT05AL	Stops the sense node from overdriving the downstream stage during spikes or abnormal excursions
ZD2	Low-voltage internal node after ZD1	Usually inactive; only conducts on transients	Clamps fast surges/ESD/RFI so downstream circuitry does not see excessive voltage
ZD3	PE-related clamp node / floating PE path	Usually inactive when PE is healthy	Limits the PE-open floating node swing; it handles polarity reversals and surges

DC-DC stepdown converters (+5V, +3V3)



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /AC power, DC power, AC sync/

File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

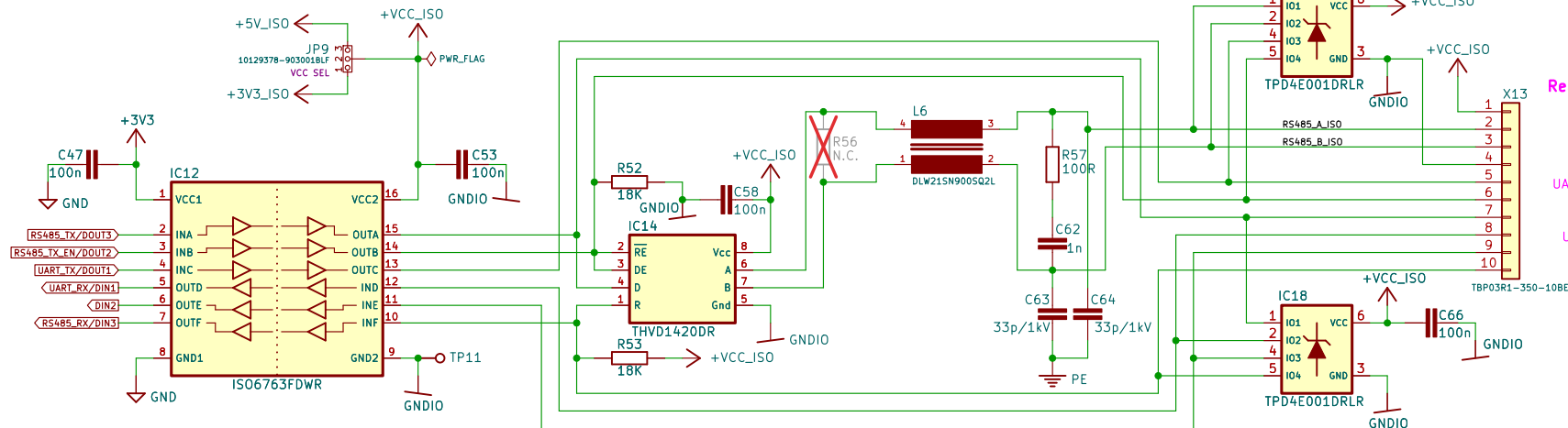
Size: A4 Date: 2026-09-19

KiCad E.D.A. 10.0.6

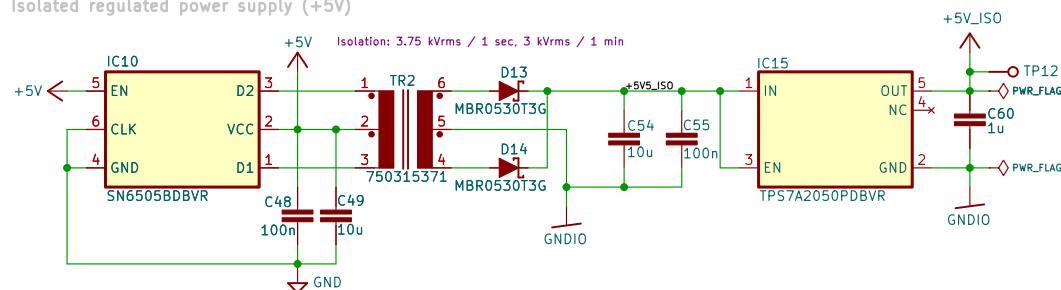
Rev: r1B3

Id: 4/10

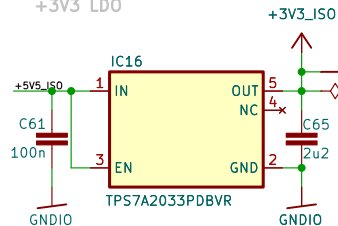
Isolated digital I/O & RS485 transceiver (half-duplex)



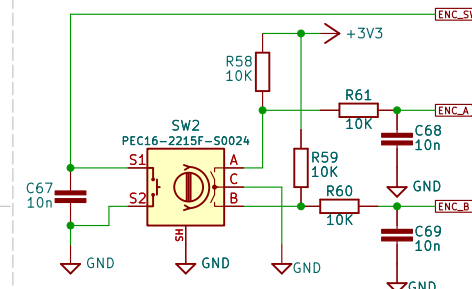
Isolated regulated power supply (+5V)



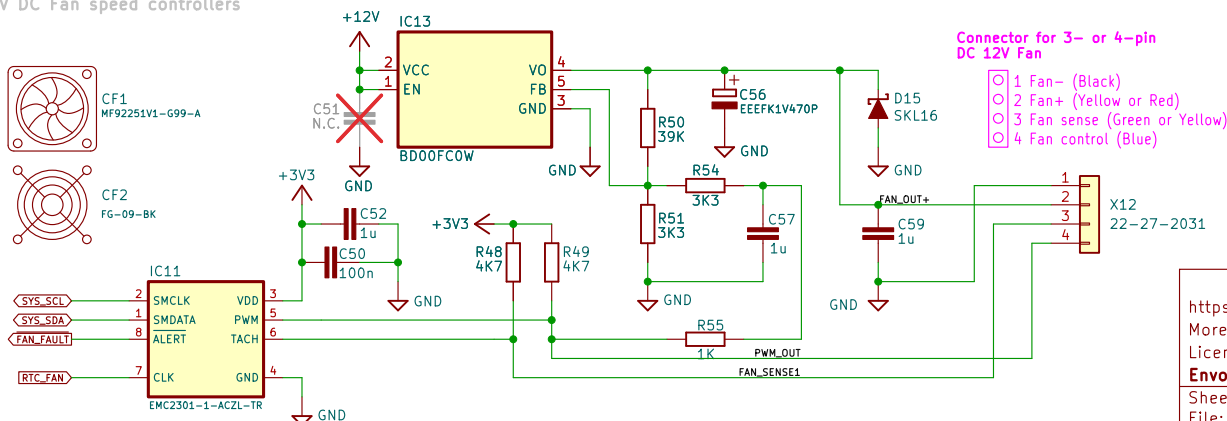
+3V3 LDO



Encoder with switch



12V DC Fan speed controllers



<https://github.com/eez-open/eez-bb3plus>

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Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

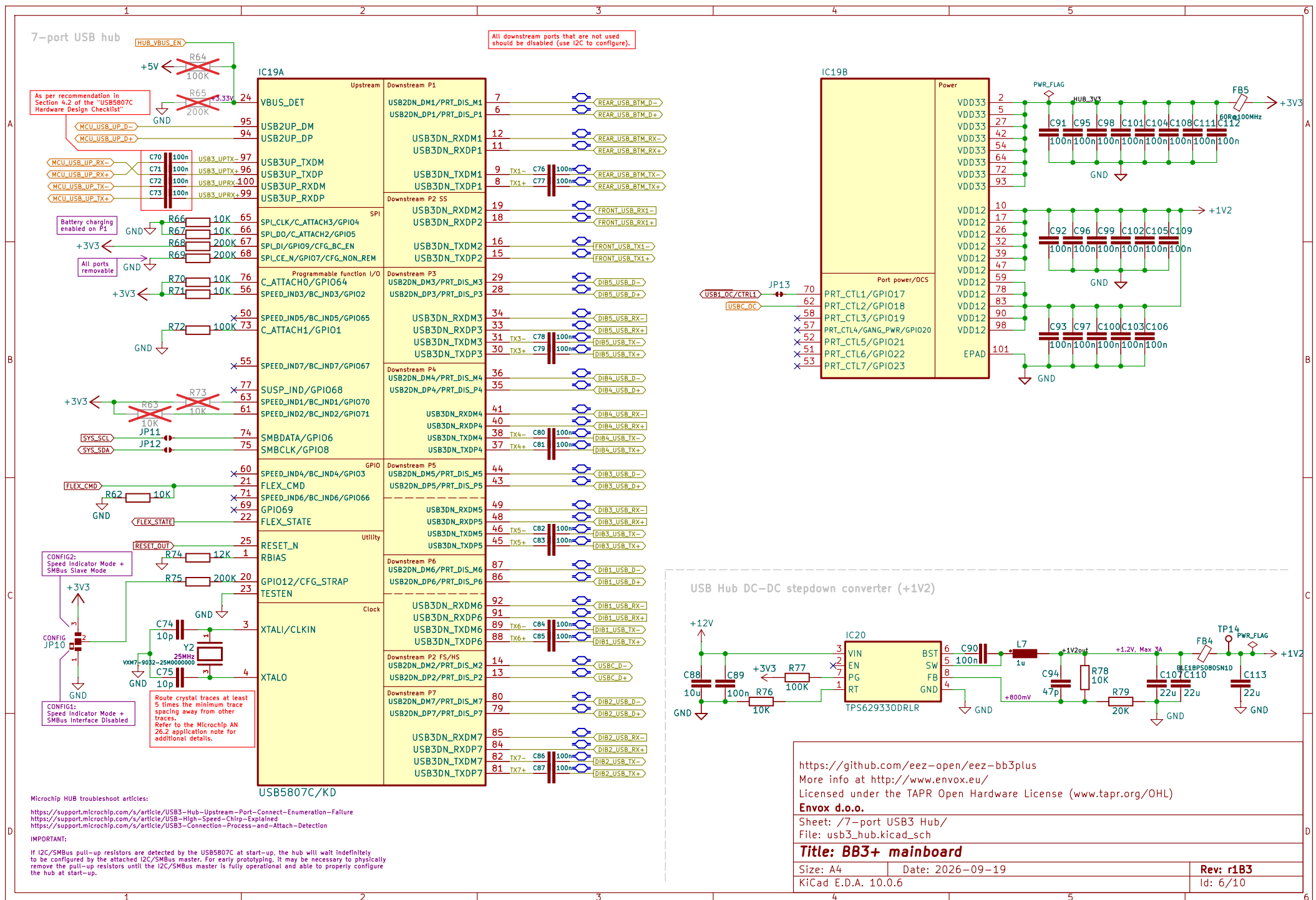
Title: BB3+ mainboard

Size: A4 Date: 2026-09-19

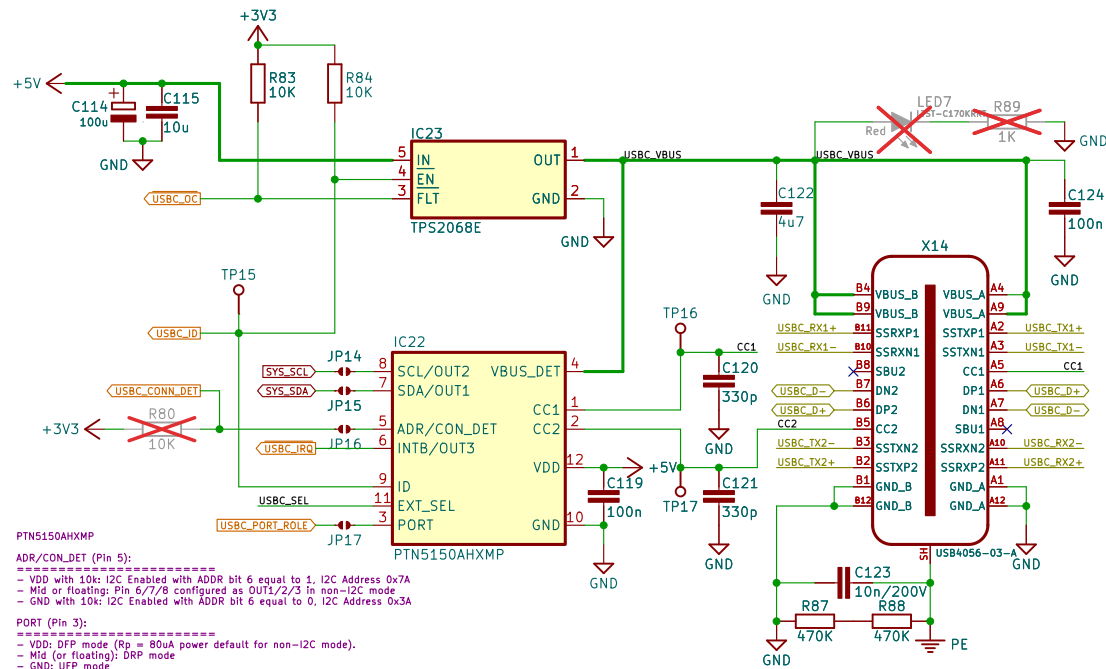
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Rev: r1B3

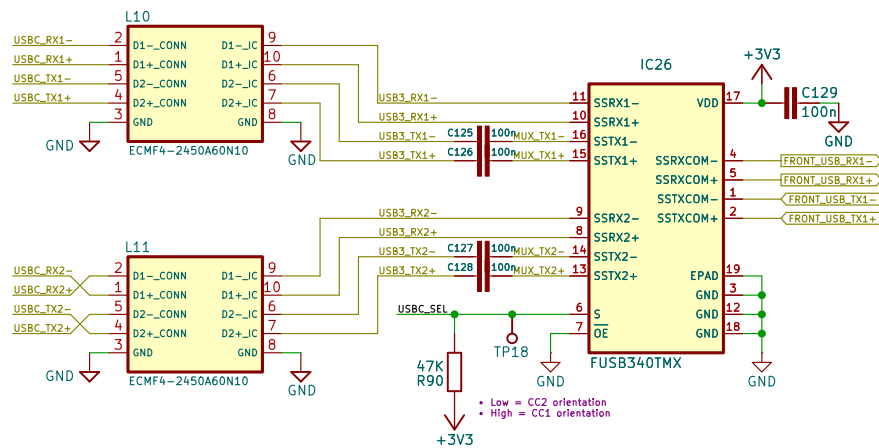
Id: 5/10



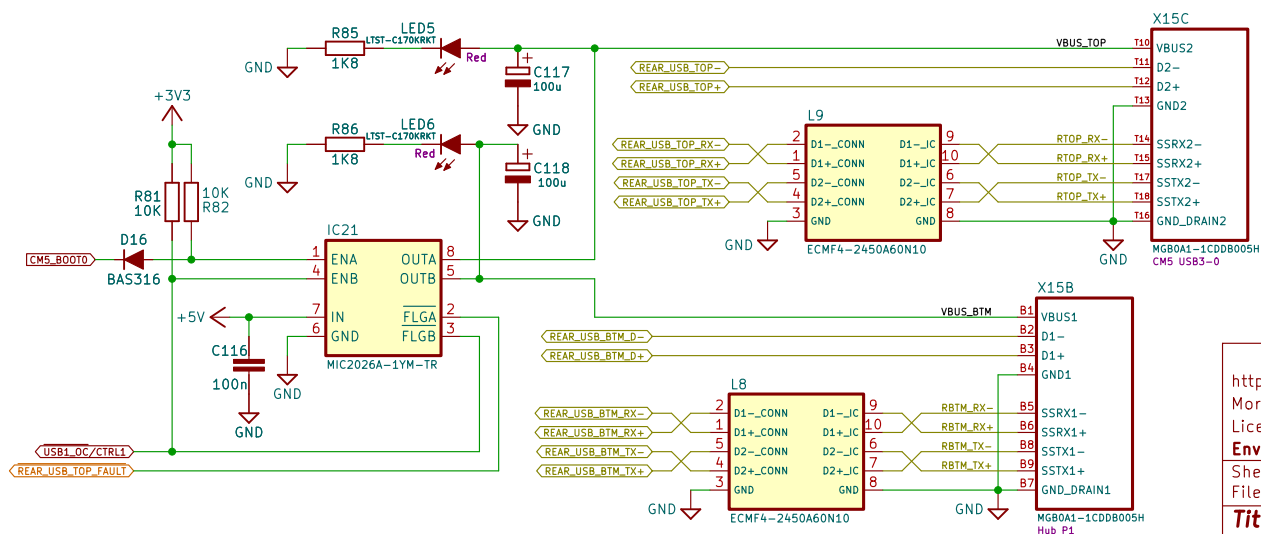
USB-C 3.0 front port (source max 1.5A)

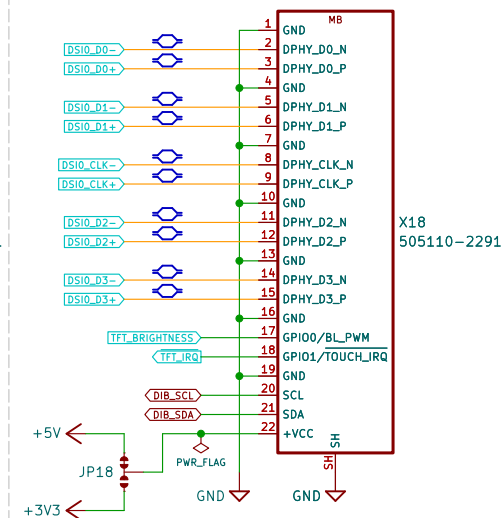
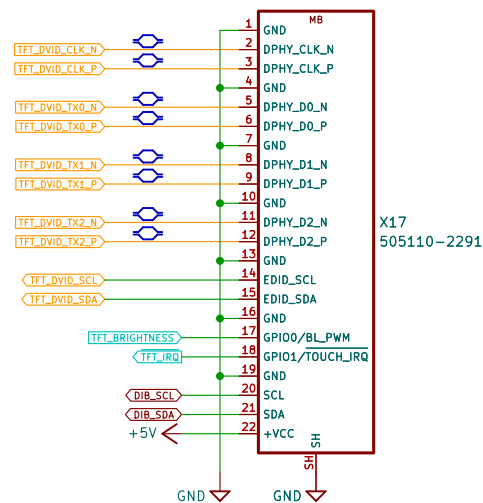
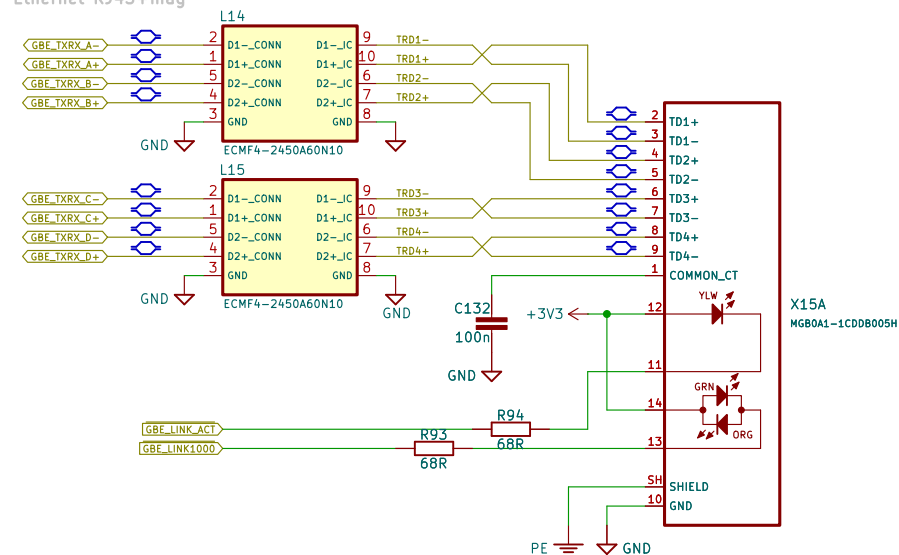
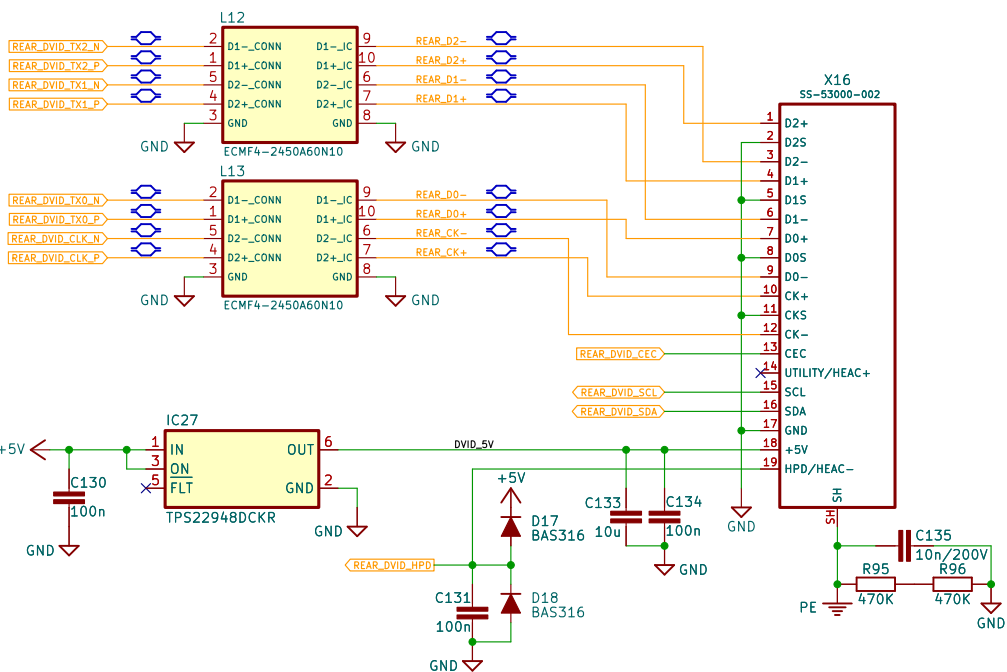


USB-C 3.0 Protection and Mux

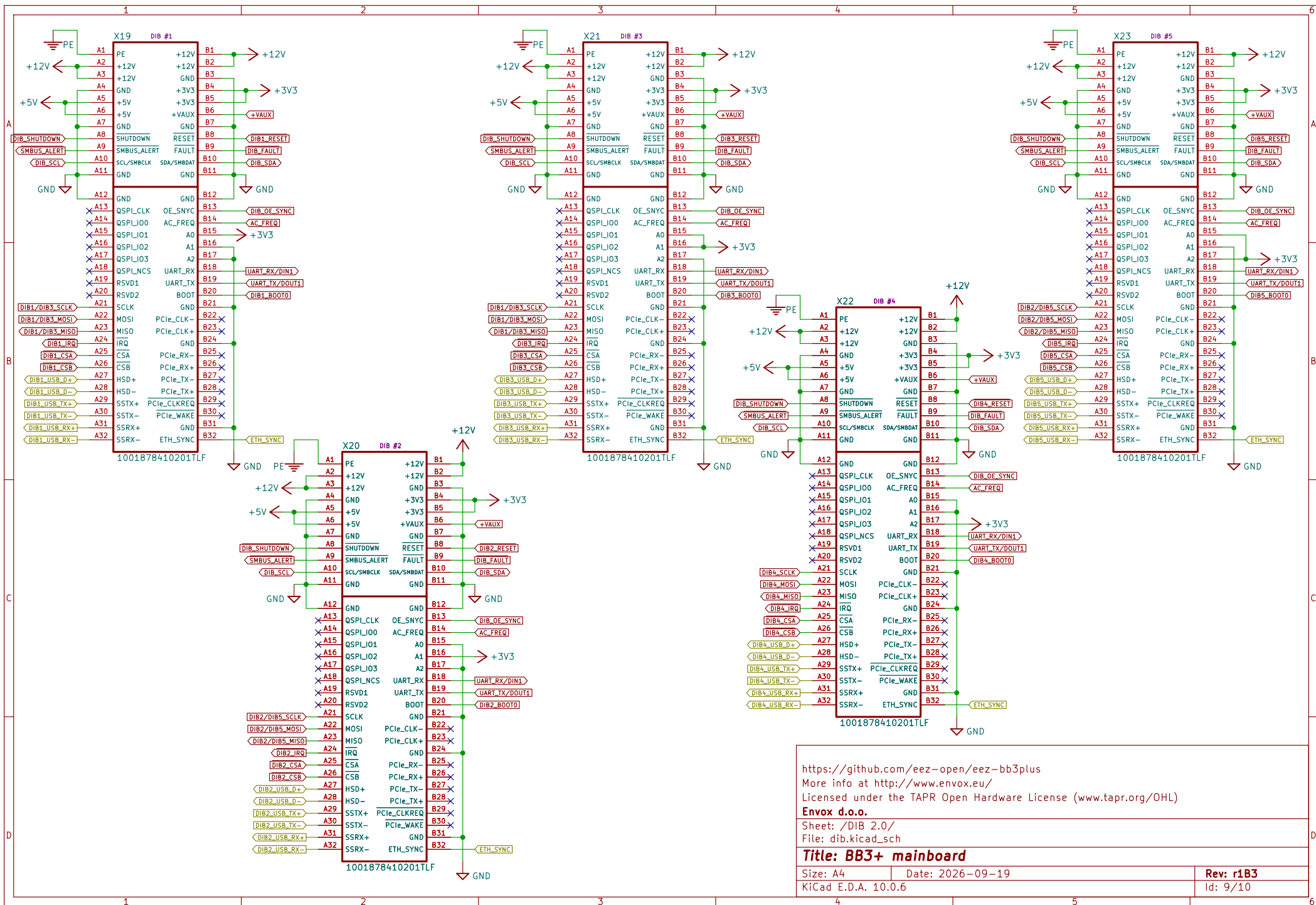


USB2.0 Type-A (Rear panel)





Id: 8/10



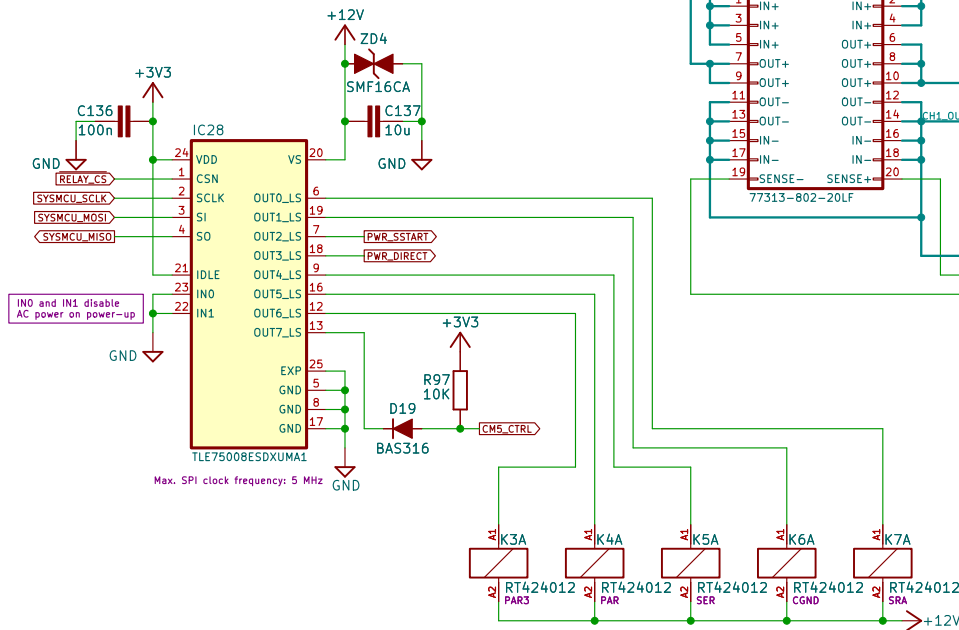
<https://github.com/eez-open/eez-bb3plus>
More info at <http://www.envox.eu/>
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Sheet: /DIB 2.0/
File: dib.kicad_sch

Title: BB3+ mainboard	
Size: A4	Date: 2026-09-19
KiCad E.D.A. 10.0.6	Rev: r1B3 Id: 9/10

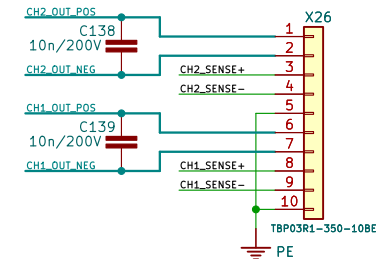
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ OUT
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 3	4
CH2_SENSE+ 5	6
CH2_SENSE- 7	8
PE 9	10
CH1_OUT+ 11	12
CH1_OUT- 13	14
CH1_SENSE+ 15	16
CH1_SENSE- 17	18
PE 19	20

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: //Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-09-19

KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 10/10